

WARRA, Australia

WMO: 959620

Lat: 43.061S

Lon: 146.704E

Elev: 1624

StdP: 13.85

Time Zone: 10.00 (AUS)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.8	32.3	28.1	23.3	35.8	29.7	25.2	36.0	27.3	46.8	24.3	45.7	4.4	260	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.7	79.3	61.4	75.0	60.2	70.9	58.9	63.6	74.6	62.0	71.4	60.2	68.6	9.2	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.1	82.4	65.1	58.4	77.3	64.5	56.6	72.6	63.4	29.8	74.6	28.5	71.3	27.3	68.7	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	20.5	18.1	DB	28.2	89.7	1.7	4.7	26.9	93.1	25.9	95.9	24.9	98.5	23.7	102.0
			WB	27.7	67.5	1.5	1.3	26.6	68.4	25.7	69.2	24.8	69.9	23.7	70.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.5	56.1	55.5	53.5	49.2	45.6	42.7	41.8	42.1	44.6	47.5	50.5	53.1	(6)
	DBStd	7.36	6.62	6.22	6.33	5.24	4.50	3.86	3.45	4.36	4.70	5.75	6.46	6.12	(7)
	HDD50	1403	15	14	32	76	150	219	256	247	173	119	68	34	(8)
	HDD65	6063	289	272	363	475	601	668	720	709	613	544	436	373	(9)
	CDD50	849	204	169	140	52	14	2	0	3	11	41	83	130	(10)
	CDD65	31	13	7	6	0	0	0	0	0	0	0	2	3	(11)
	CDH74	493	199	87	76	1	0	0	0	0	0	11	44	75	(12)
	CDH80	123	60	19	20	0	0	0	0	0	0	0	5	19	(13)
	WSAvg	8.2	8.3	7.4	7.2	7.6	8.6	7.5	8.3	8.5	8.8	8.8	8.6	8.5	(14)
	PrecAvg	55.9	3.4	2.8	3.3	3.9	5.3	5.3	6.0	7.2	6.0	5.7	4.3	4.3	(15)
(16) Precipitation	PrecMax	67.8	6.5	7.0	7.2	8.4	13.3	11.2	11.6	12.7	11.7	12.4	7.8	8.4	(16)
	PrecMin	47.2	0.6	1.2	1.5	1.5	1.9	1.7	2.3	2.4	2.4	2.0	0.9	1.5	(17)
	PrecStd	5.8	1.7	1.5	1.3	1.5	3.0	2.1	2.0	2.6	2.5	2.2	1.4	1.6	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.1	82.9	81.9	69.9	61.1	55.2	53.3	58.5	64.3	75.1	79.7	83.0	(19)
		MCWB	64.4	62.9	61.9	57.4	53.2	51.2	48.0	49.3	53.0	55.3	60.1	61.3	(20)
	2%	DB	79.1	75.9	74.2	65.6	57.3	52.5	50.7	54.4	58.9	67.6	73.4	74.5	(21)
		MCWB	61.6	61.8	60.2	56.0	51.5	48.9	46.3	47.6	49.6	54.2	58.9	59.6	(22)
	5%	DB	74.0	71.3	68.4	61.4	54.6	50.4	48.9	51.6	55.6	62.6	67.3	69.3	(23)
		MCWB	60.3	60.0	58.7	54.0	50.0	46.9	45.4	46.3	48.6	52.2	56.4	58.2	(24)
	10%	DB	68.7	67.0	64.1	57.5	52.6	48.9	47.3	49.1	52.5	58.2	62.0	64.8	(25)
		MCWB	58.3	58.4	57.1	52.3	48.9	46.0	44.5	45.2	47.1	51.0	54.2	56.0	(26)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.4	64.8	64.3	59.3	54.8	53.5	49.3	50.4	54.0	59.2	62.6	64.3	(27)
		MCDB	80.9	77.0	74.9	67.3	58.9	53.7	51.6	55.1	61.6	66.0	72.0	75.1	(28)
	2%	WB	63.7	63.1	62.0	56.8	52.3	49.4	47.1	48.7	51.3	56.0	60.3	61.4	(29)
		MCDB	75.5	73.3	70.8	63.6	55.6	51.5	49.3	53.3	56.8	64.1	70.6	71.4	(30)
	5%	WB	61.3	61.1	60.0	54.9	50.8	47.7	45.9	47.0	49.3	53.3	57.6	59.1	(31)
		MCDB	71.0	69.2	67.0	60.1	53.9	49.7	48.2	50.9	53.9	61.0	65.9	67.5	(32)
	10%	WB	58.9	59.0	57.7	52.9	49.3	46.4	44.9	45.4	47.5	51.0	54.8	56.9	(33)
		MCDB	67.6	65.9	64.2	57.1	52.3	48.5	47.1	48.6	51.8	57.5	61.4	64.1	(34)
(36) Mean Daily Temperature Range	5% DB	MDBR	18.7	17.7	15.4	12.4	9.9	9.1	9.0	10.5	12.7	15.7	16.2	17.3	(35)
		MCDBR	29.8	25.8	23.6	18.3	11.7	10.5	10.4	13.9	17.9	24.4	26.7	27.3	(36)
		MCWBR	14.3	12.4	12.4	10.7	8.5	8.2	8.0	9.1	11.1	13.6	13.6	13.8	(37)
	5% WB	MCDBR	26.1	22.9	20.9	16.7	11.2	9.3	9.8	12.9	16.0	21.6	23.5	24.0	(38)
		MCWBR	13.7	12.2	11.9	10.8	8.9	7.8	8.2	8.9	10.8	13.0	13.0	13.7	(39)
(40) Clear-Sky Solar Irradiance	taub		0.331	0.319	0.319	0.304	0.295	0.284	0.286	0.300	0.324	0.327	0.323	0.327	(40)
	taud		2.529	2.581	2.571	2.598	2.602	2.616	2.615	2.581	2.493	2.496	2.531	2.530	(41)
	Ebn at Noon		315	311	296	279	258	251	259	276	289	305	316	318	(42)
	Edh at Noon		34	31	28	23	19	17	19	23	30	33	34	35	(43)
(44) All-Sky Solar Radiation	RadAvg		1989	1711	1246	836	503	401	459	689	1038	1486	1814	1978	(44)
	RadStd		146	123	75	49	38	36	28	50	55	85	139	123	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (27 neighbors)	+0.58	N/A	N/A	+1.22	N/A	N/A	N/A	-180	+191	+25

Nomenclature: See separate page